

2023 AMC 10A Problem 19 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 19. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 19

Problem:

The line segment from $A(1, 2)$ to $B(3, 3)$ can be transformed to the line segment from $A'(3, 1)$ to $B'(4, 3)$, sending A to A' and B to B' , by a rotation centered at the point $P(s, t)$. What is $|s - t|$?

Answer Choices:

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

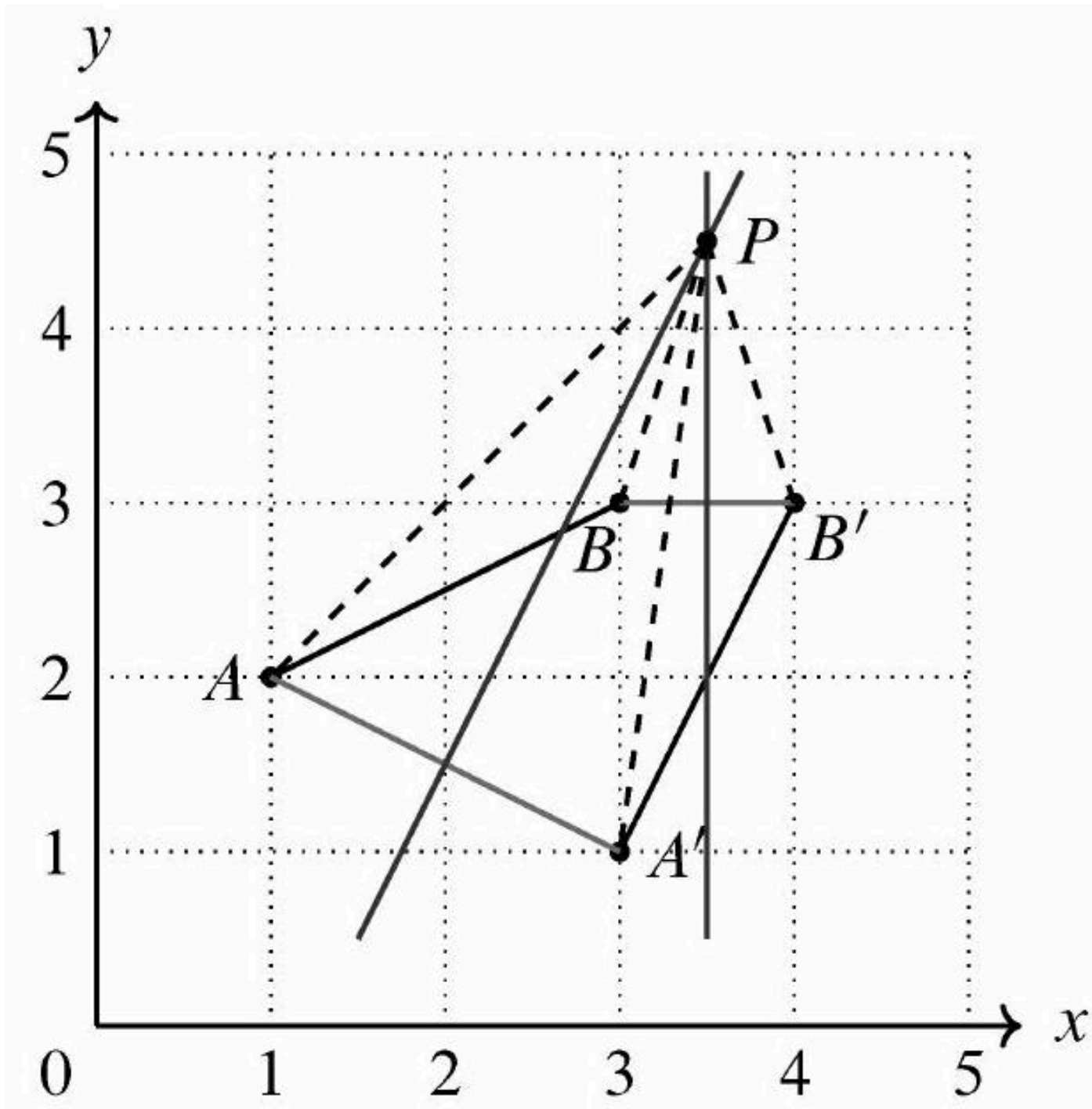
E. 1

Solution:

Because A and A' are the same distance from the center of rotation, $P(s, t)$ must lie on the perpendicular bisector of $\overline{AA'}$; similarly it must lie on the perpendicular bisector of $\overline{BB'}$. The line $\overline{AA'}$ passes through the midpoint $(2, \frac{3}{2})$ and has slope $-\frac{3-1}{1-2} = 2$, so its equation is

$$y = 2(x - 2) + \frac{3}{2}.$$

The line $\overline{BB'}$ is vertical and passes through $(\frac{7}{2}, 3)$, so its equation is $x = \frac{7}{2}$. Solving this system of equations gives $P(s, t) = (\frac{7}{2}, \frac{9}{2})$, as shown in the figure. The requested absolute difference of coordinates is $|\frac{7}{2} - \frac{9}{2}| = (E)\boxed{1}$.



Note: As a check, note that the angle of rotation must be $\angle APA'$ and also $\angle BPB'$; it is about 37° . By the SSS Congruence Theorem, $\triangle APB \cong \triangle A'PB'$, so indeed

$$\angle APA' = \angle APB + \angle BPA' = \angle A'PB' + \angle BPA' = \angle BPB'.$$

Thus the rotation of this amount around $\left(\frac{7}{2}, \frac{9}{2}\right)$ is the required transformation.

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